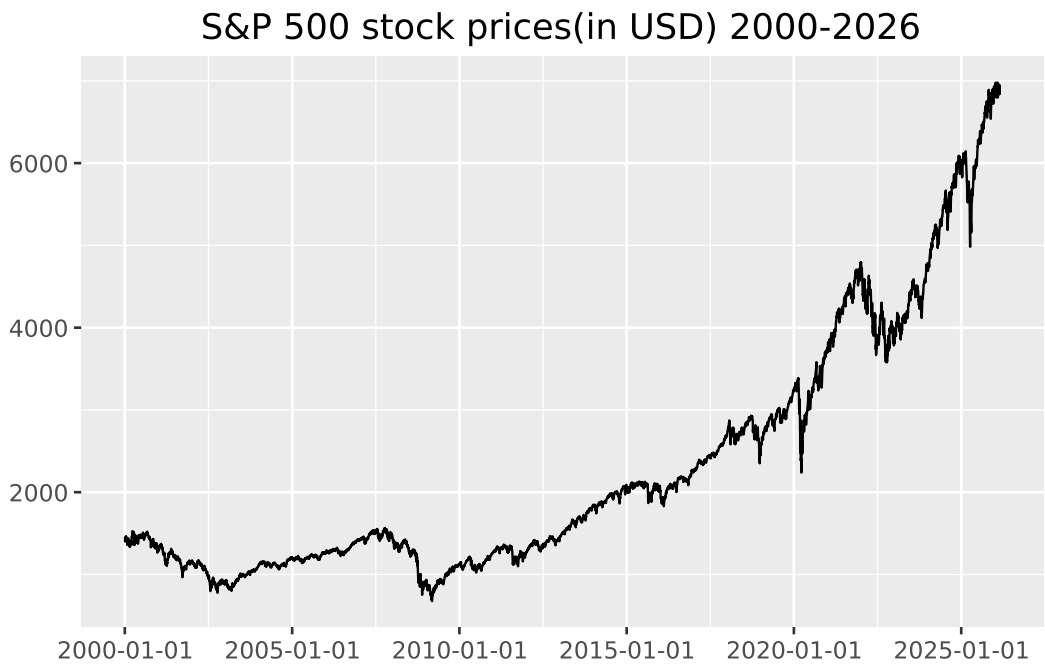
Mandatory Assignment 1

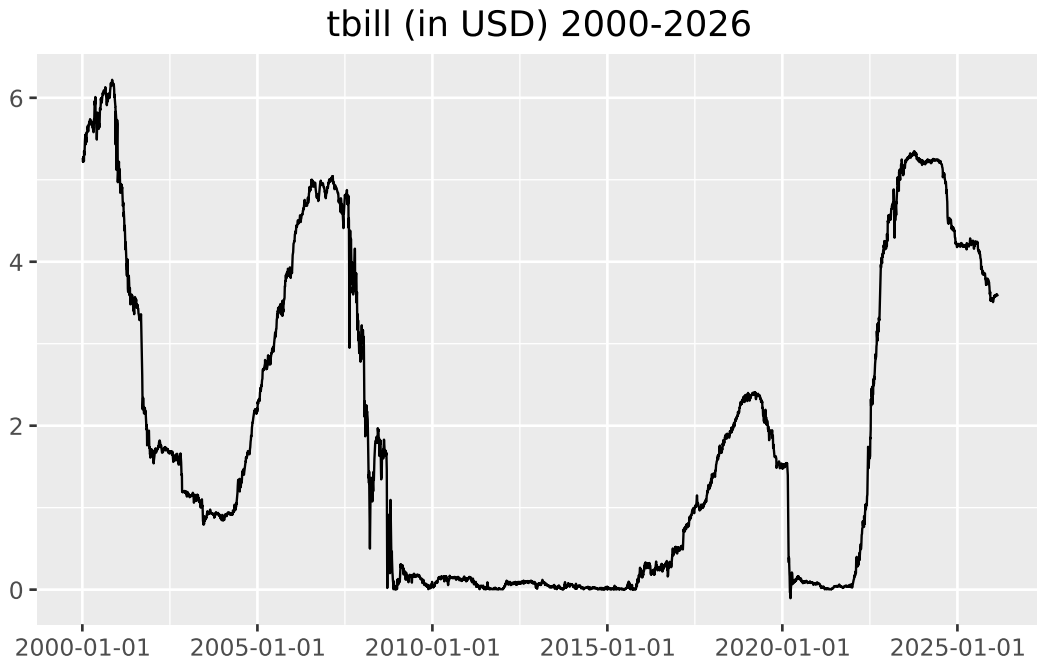
GFS189, RTL959

March 06, 2026

.

1 Exercise 1





1. Visualize the distribution of the daily returns.

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Monthly mean excess return:       0.006
Monthly variance:                 0.002
Annualized mean excess return:    0.071
Annualized variance:              0.023
```

1.1 Exercise 2

1.2 Exercise 3

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Tangency portfolio Sharpe: 0.1335
Tangency mu (monthly):     0.0063
Tangency sigma (monthly):  0.0473
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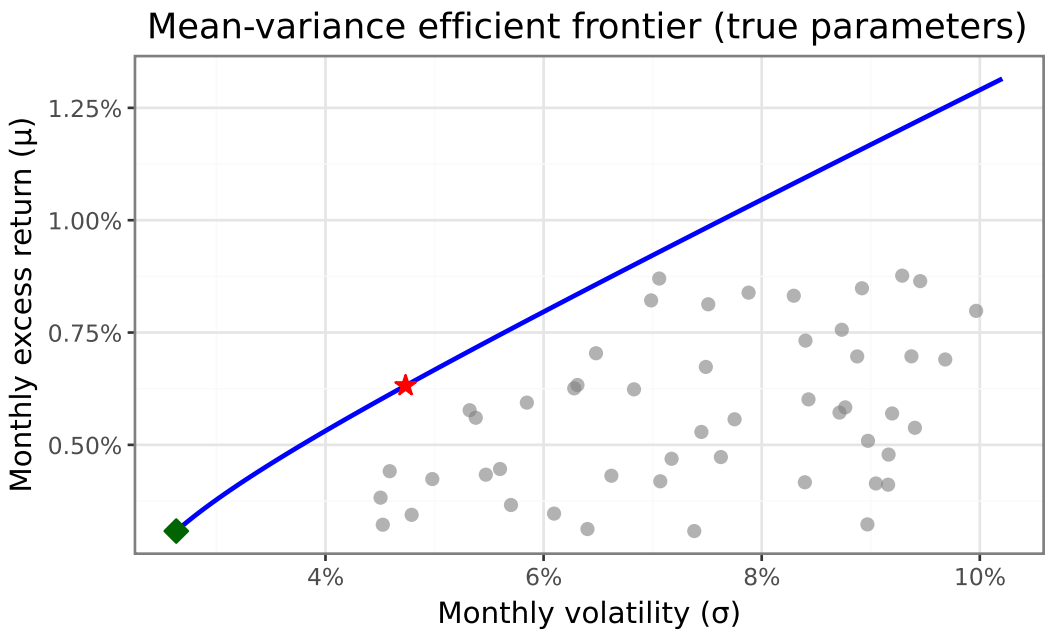
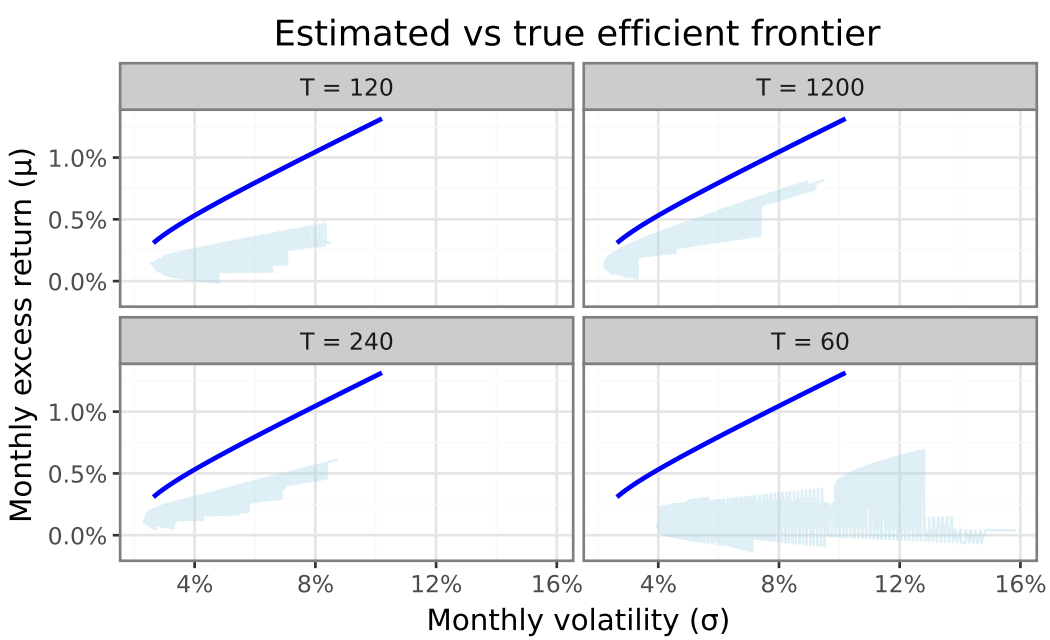
1.3 Exercise 4



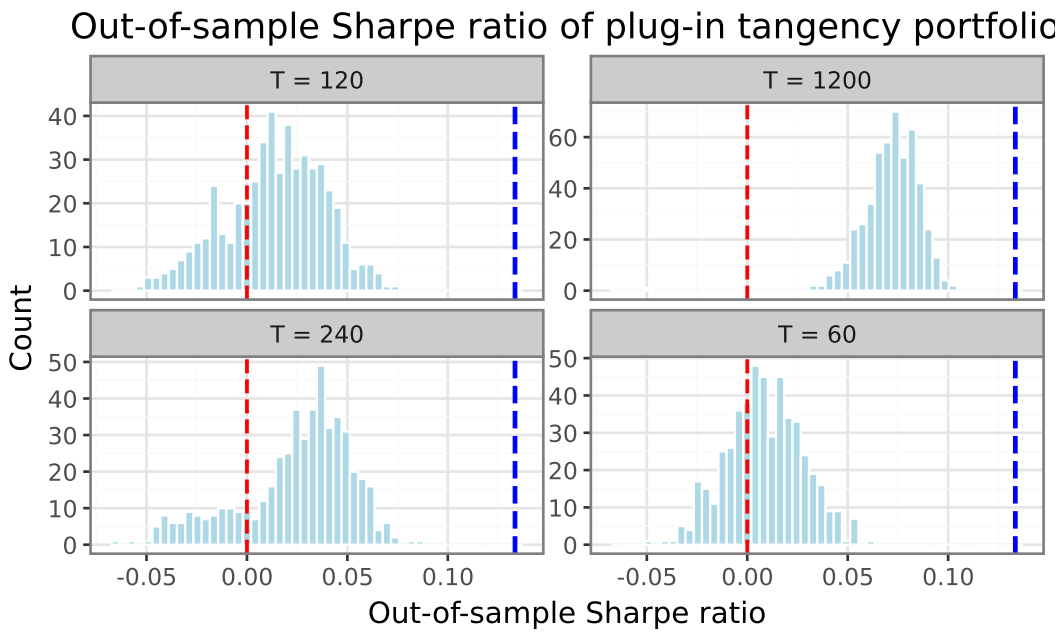
gency portfolio. Green diamond: minimum variance portfolio. Grey dots: individual assets.

1.4 Exercise 5

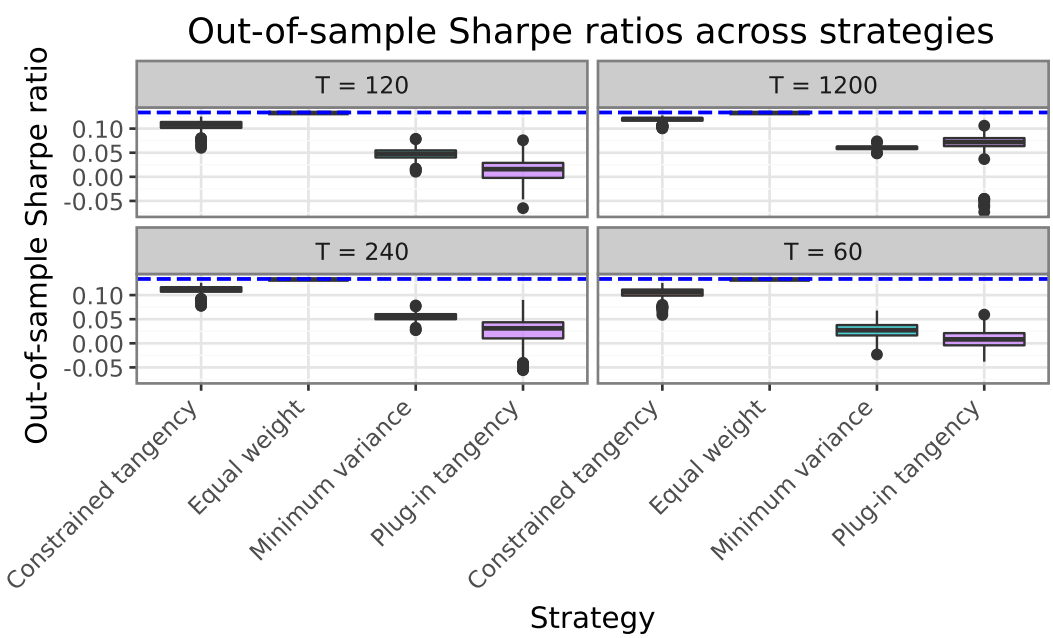


Light blue: plug-in frontiers. Dark blue: true frontier.

1.5 Exercise 6



Dashed blue: true tangency Sharpe. Dashed red: zero.



Dashed blue: true tangency Sharpe.

References